Model for "Te"

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General Condition

- Basis type: `lgs`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 152 D_3^4 $P3_121$ [trigonal]
- Associated point group: PG No. 152 D_3 32 (321 setting) [trigonal]
- Unit cell:
  $a =$ `4.45000`, $b =$ `4.45000`, $c =$ `5.93000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `120.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 4.45000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[-2.22500, 3.85381, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 5.93000]`

Symmetry Operation

Table 1: Symmetry operation

	#	SO	#	SO	#	SO	#	SO	#	SO
	1	$\{1\|0\}$	2	$\{3^{+}_{001}\|00\frac{1}{3}\}$	3	$\{3^{-}_{001}\|00\frac{2}{3}\}$	4	$\{2_{110}\|0\}$	5	$\{2_{100}\|00\frac{2}{3}\}$
	6	$\{2_{010}\|00\frac{1}{3}\}$								

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{G}_0(A_1)$	A_1	0	G, M	-	-	1
2	$\mathbb{Q}_0(A_1)$	A_1	0	Q, T	-	-	1
3	$\mathbb{G}_2(A_1)$	A_1	2	G, M	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
4	$\mathbb{Q}_2(A_1)$	A_1	2	Q, T	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
5	$\mathbb{G}_3(A_1)$	A_1	3	G, M	-	-	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
6	$\mathbb{Q}_3(A_1)$	A_1	3	Q, T	-	-	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
7	$\mathbb{Q}_4(A_1,2)$	A_1	4	Q, T	2	-	$\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$
8	$\mathbb{G}_1(A_2)$	A_2	1	G, M	-	-	z
9	$\mathbb{Q}_1(A_2)$	A_2	1	Q, T	-	-	z

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
10	$\mathbb{G}_3(A_2,1)$	A_2	3	G, M	1	-	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$
11	$\mathbb{G}_3(A_2,2)$	A_2	3	G, M	2	-	$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$
12	$\mathbb{G}_{1,1}(E)$	E	1	G, M	-	1	x
13	$\mathbb{G}_{1,2}(E)$					2	y
14	$\mathbb{Q}_{1,1}(E)$	E	1	Q, T	-	1	x
15	$\mathbb{Q}_{1,2}(E)$					2	y
16	$\mathbb{Q}_{2,1}(E,1)$	E	2	Q, T	1	1	$\sqrt{3}yz$
17	$\mathbb{Q}_{2,2}(E,1)$					2	$-\sqrt{3}xz$
18	$\mathbb{Q}_{2,1}(E,2)$	E	2	Q, T	2	1	$\frac{\sqrt{3}(x^2-y^2)}{2}$
19	$\mathbb{Q}_{2,2}(E,2)$					2	$-\sqrt{3}xy$
20	$\mathbb{G}_{3,1}(E,1)$	E	3	G, M	1	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
21	$\mathbb{G}_{3,2}(E,1)$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
22	$\mathbb{G}_{3,1}(E,2)$	E	3	G, M	2	1	$\sqrt{15}xyz$
23	$\mathbb{G}_{3,2}(E,2)$					2	$\frac{\sqrt{15}z(x^2-y^2)}{2}$

Basis in full matrix

Table 3: dimension = 18

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	0	$\|p_x,\uparrow\rangle$ @Te(1)	1	$\|p_x,\downarrow\rangle$ @Te(1)	2	$\|p_y,\uparrow\rangle$ @Te(1)	3	$\|p_y,\downarrow\rangle$ @Te(1)	4	$\|p_z,\uparrow\rangle$ @Te(1)
	5	$\|p_z,\downarrow\rangle$ @Te(1)	6	$\|p_x,\uparrow\rangle$ @Te(2)	7	$\|p_x,\downarrow\rangle$ @Te(2)	8	$\|p_y,\uparrow\rangle$ @Te(2)	9	$\|p_y,\downarrow\rangle$ @Te(2)
	10	$\|p_z,\uparrow\rangle$ @Te(2)	11	$\|p_z,\downarrow\rangle$ @Te(2)	12	$\|p_x,\uparrow\rangle$ @Te(3)	13	$\|p_x,\downarrow\rangle$ @Te(3)	14	$\|p_y,\uparrow\rangle$ @Te(3)
	15	$\|p_y,\downarrow\rangle$ @Te(3)	16	$\|p_z,\uparrow\rangle$ @Te(3)	17	$\|p_z,\downarrow\rangle$ @Te(3)				

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

SAMB: 66 (all 66)

- `Te` : 'Te' site-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a`

  z1 $\quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

  z2 $\quad \mathbb{Q}_2^{(c)}(A_1) = \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

z3 $\mathbb{Q}_3^{(c)}(A_1) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)\right)}{2}$

z4 $\mathbb{Q}_2^{(1,-1;c)}(A_1) = \mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

z5 $\mathbb{Q}_3^{(1,-1;c)}(A_1) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)\right)}{2}$

z6 $\mathbb{Q}_0^{(1,1;c)}(A_1) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

z7 $\mathbb{G}_2^{(c)}(A_1) = -\dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)\right)}{2}$

z8 $\mathbb{G}_2^{(1,-1;c)}(A_1) = -\dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)\right)}{2}$

z9 $\mathbb{G}_0^{(1,0;c)}(A_1) = \dfrac{\sqrt{2}\left(\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)\right)}{2}$

- `Te;Te_001_1` : 'Te'-'Te' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@3b`

z10 $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

z11 $\mathbb{Q}_2^{(c)}(A_1) = \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

z12 $\mathbb{Q}_3^{(c)}(A_1) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z13 $\mathbb{Q}_2^{(1,-1;c)}(A_1) = \mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

z14 $\mathbb{Q}_3^{(1,-1;c)}(A_1,a) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z15 $\mathbb{Q}_3^{(1,-1;c)}(A_1,b) = \mathbb{M}_3^{(1,-1;a)}(A_2,2)\mathbb{T}_1^{(b)}(A_2)$

z16 $$\mathbb{Q}_{3}^{(1,-1;c)}(A_1,c) = -\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z17 $$\mathbb{Q}_{3}^{(1,0;c)}(A_1) = \frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z18 $$\mathbb{Q}_{0}^{(1,1;c)}(A_1) = \mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$$

z19 $$\mathbb{G}_{0}^{(c)}(A_1,a) = \mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$$

z20 $$\mathbb{G}_{0}^{(c)}(A_1,b) = \frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z21 $$\mathbb{G}_{2}^{(c)}(A_1) = -\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$$

z22 $$\mathbb{G}_{0}^{(1,-1;c)}(A_1,a) = \mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$$

z23 $$\mathbb{G}_{0}^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z24 $$\mathbb{G}_{2}^{(1,-1;c)}(A_1,a) = -\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$$

z25 $$\mathbb{G}_{2}^{(1,-1;c)}(A_1,b) = \mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)$$

z26 $$\mathbb{G}_{2}^{(1,-1;c)}(A_1,c) = \frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z27 $$\mathbb{G}_{0}^{(1,0;c)}(A_1) = \frac{\sqrt{2}\left(\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$$

z28 $$\mathbb{G}_{2}^{(1,0;c)}(A_1) = -\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E,a) + \mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$$

z29 $$\mathbb{G}_{0}^{(1,1;c)}(A_1,a) = \mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$$

z30 $\mathbb{G}_{0}^{(1,1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

- `Te;Te_002_1` : 'Te'-'Te' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6a@6c`

z31 $\mathbb{Q}_{0}^{(c)}(A_1,a)=\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

z32 $\mathbb{Q}_{0}^{(c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)+\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)\right)}{2}$

z33 $\mathbb{Q}_{0}^{(c)}(A_1,c)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{M}_{1,2}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z34 $\mathbb{Q}_{2}^{(c)}(A_1)=\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

z35 $\mathbb{Q}_{3}^{(c)}(A_1)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)+\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z36 $\mathbb{Q}_{4}^{(c)}(A_1,2)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)+\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)\right)}{2}$

z37 $\mathbb{Q}_{0}^{(1,-1;c)}(A_1,a)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)+\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)\right)}{2}$

z38 $\mathbb{Q}_{0}^{(1,-1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z39 $\mathbb{Q}_{2}^{(1,-1;c)}(A_1,a)=\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

z40 $\mathbb{Q}_{2}^{(1,-1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z41 $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,a)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)+\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z42 $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,b)=\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1}^{(b)}(A_2)$

z43 $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,c)=-\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z44 $\mathbb{Q}_{4}^{(1,-1;c)}(A_1,2a)=\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)+\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)\right)}{2}$

z45 $\mathbb{Q}_{4}^{(1,-1;c)}(A_1,2b)=\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z46 $\mathbb{Q}_{2}^{(1,0;c)}(A_1,a)=\frac{\sqrt{2}\left(\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)+\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)\right)}{2}$

z47 $\mathbb{Q}_{2}^{(1,0;c)}(A_1,b)=\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{0}^{(b)}(A_1)$

z48 $\mathbb{Q}_{2}^{(1,0;c)}(A_1,c)=-\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z49 $\mathbb{Q}_{3}^{(1,0;c)}(A_1)=\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z50 $\mathbb{Q}_{0}^{(1,1;c)}(A_1,a)=\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

z51 $\mathbb{Q}_{0}^{(1,1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z52 $\mathbb{G}_{0}^{(c)}(A_1,a)=\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$

z53 $\mathbb{G}_{0}^{(c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z54 $\mathbb{G}_{2}^{(c)}(A_1)=-\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)+\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z55 $\mathbb{G}_{0}^{(1,-1;c)}(A_1,a)=\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$

z56 $\mathbb{G}_{0}^{(1,-1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z57 $\mathbb{G}_{2}^{(1,-1;c)}(A_1,a)=-\frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)+\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)\right)}{2}$

z58 $\mathbb{G}_{2}^{(1,-1;c)}(A_1,b)=\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)$

z59 $\mathbb{G}_{2}^{(1,-1;c)}(A_1,c)=\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z60 $\mathbb{G}_{3}^{(1,-1;c)}(A_1)=\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{0}^{(b)}(A_1)$

z61 $\mathbb{G}_{0}^{(1,0;c)}(A_1)=\frac{\sqrt{3}\left(\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)+\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)+\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)\right)}{3}$

z62 $\mathbb{G}_{2}^{(1,0;c)}(A_1,a)=\frac{\sqrt{6}\left(-\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)-\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)+2\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)\right)}{6}$

z63 $\mathbb{G}_{2}^{(1,0;c)}(A_1,b)=-\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

z64 $\mathbb{G}_{3}^{(1,0;c)}(A_1)=\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)+\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)\right)}{2}$

z65 $\mathbb{G}_{0}^{(1,1;c)}(A_1,a)=\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)$

z66 $\mathbb{G}_{0}^{(1,1;c)}(A_1,b)=\frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E,a)+\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E,a)\right)}{2}$

Atomic SAMB

- bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
- ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$

x1 $\mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$

x2 $\mathbb{Q}_2^{(a)}(A_1) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x3 $\mathbb{Q}_{2,1}^{(a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

x4 $\mathbb{Q}_{2,2}^{(a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x5 $\mathbb{Q}_{2,1}^{(a)}(E,2) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x6 $\mathbb{Q}_{2,2}^{(a)}(E,2) = \begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x7 $\mathbb{Q}_{2}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

x8 $\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

x9 $\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

x10 $\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

x11 $\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x12 $\mathbb{Q}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

x13 $\mathbb{G}_1^{(1,0;a)}(A_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x14 $\mathbb{G}_{1,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

x15 $\mathbb{G}_{1,2}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

x16 $\mathbb{M}_1^{(a)}(A_2) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x17 $\mathbb{M}_{1,1}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

x18 $\mathbb{M}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x19 $\mathbb{M}_{3}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x20 $\mathbb{M}_{1}^{(1,-1;a)}(A_2) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

x21 $\mathbb{M}_3^{(1,-1;a)}(A_2,1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$

x22 $\mathbb{M}_3^{(1,-1;a)}(A_2,2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x23 $\mathbb{M}_{1,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

x24 $\mathbb{M}_{1,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

x25 $\mathbb{M}_{3,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{60} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \\ 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & \frac{\sqrt{30}}{15} & 0 \end{bmatrix}$

x26 $\mathbb{M}_{3,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & \frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$

x27 $\mathbb{M}_{3,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$

x28 $\mathbb{M}_{3,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

x29 $\mathbb{M}_1^{(1,1;a)}(A_2) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$

x30 $\mathbb{M}_{1,1}^{(1,1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$

x31 $\mathbb{M}_{1,2}^{(1,1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$

x32 $\mathbb{T}_2^{(1,0;a)}(A_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x33 $\mathbb{T}_{2,1}^{(1,0;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

x34 $\mathbb{T}_{2,2}^{(1,0;a)}(E,1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

x35 $\mathbb{T}_{2,1}^{(1,0;a)}(E,2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$

x36 $\mathbb{T}_{2,2}^{(1,0;a)}(E,2) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

- Site cluster

** Wyckoff: `3a`

y1 $\mathbb{Q}_{0}^{(s)}(A_1) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y2 $\mathbb{Q}_{1,1}^{(s)}(E) = \left[\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y3 $\mathbb{Q}_{1,2}^{(s)}(E) = \left[0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

- Bond cluster

** Wyckoff: `3a@3b`

y4 $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y5 $\mathbb{T}_{1}^{(b)}(A_2) = \left[\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}\right]$

y6 $\mathbb{Q}_{1,1}^{(b)}(E) = \left[\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y7 $\mathbb{Q}_{1,2}^{(b)}(E) = \left[0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

y8 $\mathbb{T}_{1,1}^{(b)}(E, a) = \left[0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2}\right]$

y9 $\mathbb{T}_{1,2}^{(b)}(E, a) = \left[\frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

** Wyckoff: `6a@6c`

y10 $\mathbb{Q}_0^{(b)}(A_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

y11 $\mathbb{T}_0^{(b)}(A_1) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$

y12 $\mathbb{Q}_1^{(b)}(A_2) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y13 $\mathbb{T}_1^{(b)}(A_2) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

y14 $\mathbb{Q}_{1,1}^{(b)}(E) = \left[\frac{2\sqrt{21}}{21}, -\frac{5\sqrt{21}}{42}, \frac{\sqrt{21}}{42}, \frac{\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, -\frac{5\sqrt{21}}{42}\right]$

y15 $\mathbb{Q}_{1,2}^{(b)}(E) = \left[\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, \frac{3\sqrt{7}}{14}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{14}\right]$

y16 $\mathbb{T}_{1,1}^{(b)}(E, a) = \left[-\frac{5\sqrt{21}i}{42}, \frac{\sqrt{21}i}{42}, \frac{2\sqrt{21}i}{21}, \frac{2\sqrt{21}i}{21}, -\frac{5\sqrt{21}i}{42}, \frac{\sqrt{21}i}{42}\right]$

y17 $\mathbb{T}_{1,2}^{(b)}(E, a) = \left[\frac{\sqrt{7}i}{14}, -\frac{3\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{14}, \frac{3\sqrt{7}i}{14}\right]$

y18 $\mathbb{M}_{1,1}^{(b)}(E) = \left[\frac{\sqrt{7}i}{14}, -\frac{3\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{14}, -\frac{3\sqrt{7}i}{14}\right]$

y19 $\mathbb{M}_{1,2}^{(b)}(E) = \left[\frac{5\sqrt{21}i}{42}, -\frac{\sqrt{21}i}{42}, -\frac{2\sqrt{21}i}{21}, \frac{2\sqrt{21}i}{21}, -\frac{5\sqrt{21}i}{42}, \frac{\sqrt{21}i}{42}\right]$

y20 $\mathbb{Q}_{2,1}^{(b)}(E, 1) = \left[\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{14}\right]$

y21 $\mathbb{Q}_{2,2}^{(b)}(E, 1) = \left[-\frac{2\sqrt{21}}{21}, \frac{5\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, -\frac{5\sqrt{21}}{42}\right]$

Table 5: Orbital of each site

#	site	orbital
1	Te	$\|p_x,\uparrow\rangle$, $\|p_x,\downarrow\rangle$, $\|p_y,\uparrow\rangle$, $\|p_y,\downarrow\rangle$, $\|p_z,\uparrow\rangle$, $\|p_z,\downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	Te	Te	[1,2]	[p]	[p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Te' (#1) site cluster (3a), .2.

SL	position ($\boldsymbol{s}$)	mapping
1	[0.26360, 0.00000, 0.33333]	[1,5]
2	[0.00000, 0.26360, 0.66667]	[2,4]
3	[0.73640, 0.73640, 0.00000]	[3,6]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'Te'-'Te' [1] (#1) bond cluster (3a@3b), ND, $|\boldsymbol{v}|$= 2.83463 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.26360,-0.52720, 0.33333]	[0.86820, 0.00000, 0.83333]	[1,-5]	(3,1)	(2,1)	[1,1,-1]
2	[0.52720, 0.26360, 0.33333]	[0.00000, 0.86820, 0.16667]	[2,-4]	(1,1)	(3,1)	[-1,-1,0]
3	[-0.26360, 0.26360, 0.33333]	[0.13180, 0.13180, 0.50000]	[3,-6]	(2,1)	(1,1)	[0,0,0]

Table 9: 2-th 'Te'-'Te' [1] (#2) bond cluster (6a@6c), D, $|\boldsymbol{v}|$= 3.48967 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.26360,-0.73640, 0.33333]	[0.13180, 0.63180, 0.50000]	[1]	(2,1)	(1,1)	[0,1,0]
2	[0.73640, 0.47280, 0.33333]	[0.36820, 0.50000, 0.83333]	[2]	(3,1)	(2,1)	[0,0,-1]
3	[-0.47280, 0.26360, 0.33333]	[0.50000, 0.86820, 0.16667]	[3]	(1,1)	(3,1)	[0,-1,0]
4	[-0.73640,-0.26360,-0.33333]	[0.63180, 0.13180, 0.50000]	[4]	(1,1)	(2,1)	[1,0,0]
5	[0.47280, 0.73640,-0.33333]	[0.50000, 0.36820, 0.16667]	[5]	(3,1)	(1,1)	[0,0,0]
6	[0.26360,-0.47280,-0.33333]	[0.86820, 0.50000, 0.83333]	[6]	(2,1)	(3,1)	[-1,0,1]